

Weekly Coding and Implementation Progress Report

First Six Weeks of Development

April 21, 2026

Overview

This report summarizes the first six weeks of development on the MFJA third-floor Gazebo project. During this period, the work evolved from basic repository setup and map reconstruction into a structured simulation platform that includes the environment model, multi-robot architecture, and the first working shuttle implementation.

The main progression across these six weeks can be described in three stages:

- building and correcting the simulation environment;
- organizing the repository and multi-robot runtime architecture;
- implementing the first practical shuttle and switch behavior.

Six-Week Summary

Week	Main Focus	Main Result
1	Repository setup and map reconstruction	Established the base workflow by testing the main repository, studying multi-robot references, and restarting the CAD-based reconstruction of the MFJA map.
2	Gazebo environment usability	Integrated the floor mesh and corrected practical environment issues such as mirrored geometry, doors, room details, windows, and STL orientation.
3	Codebase organization and collaboration workflow	Worked across branches and pull requests, and improved instructions and documentation to make the changes easier to test and review.
4	Conveyor and rail environment integration	Added more real lab content, exported static parts into Gazebo-ready meshes, updated the map accuracy, and pushed a more realistic version to the main branch.
5	ROS-Gazebo software architecture for robots	Built the launch and spawning structure for world loading, robot filtering, robot state publishing, bridge generation, and namespaced robot support.
6	Shuttle and switch implementation	Explored dynamic shuttle behavior, identified its limitations, and replaced it with a first useful kinematic shuttle implementation based on spline paths and position control.

Week 1

During Week 1, I began by setting up and testing the main simulation repository for the MFJA third-floor project. I switched to the new `mfja_3rd_floor_gz` repository and reviewed the `tb3_multi_robot` project as a reference for multi-robot spawning in Gazebo Harmonic. This helped me understand how launch files build robot creation commands and how spawning workflows are structured in practice.

In parallel, I started rebuilding the MFJA map in Autodesk Inventor. This included adding the missing room and preparing the CAD-to-Gazebo workflow so the environment could later be transferred into the simulation in a controlled and reusable way.

Week 2

During Week 2, I focused on making the simulation environment usable in practice, not only visually correct. I exported the floor mesh as STL, integrated it into the Gazebo model structure, and worked on the key model files such as `mfja_3rd_floor.sdf` and `model.config`.

I also fixed several concrete issues in the environment:

- mirrored geometry;
- incorrect door dimensions;
- missing room details;
- interior window integration for Room 315;
- STL orientation problems.

This week was important because it turned the environment into something that could actually support later robotic testing rather than remaining only a rough visual mockup.

Week 3

During Week 3, the work moved from only editing the environment to organizing the codebase and development workflow itself. I worked on several branches, opened pull requests, and started preparing my changes to be merged into the AIP-Primeca repository main branch.

At this stage, I also started writing clearer test instructions and documentation so that the changes, especially those related to multi-robot support, could be understood, tested, and reviewed more easily by others. This made the project more maintainable and more collaborative.

Week 4

During Week 4, I integrated more real lab content into the simulation. I received the conveyor and rail parts, identified which components were static, and exported all static parts into a single STL file for Gazebo integration.

I also updated the map to a newer version with more accurate windows and additional room details, pushed the current version to the main branch, and added a detailed `README`. This was the week where the environment became much closer to the real MFJA layout and started looking like a real digital counterpart of the physical lab.

Week 5

During Week 5, I worked on the actual ROS–Gazebo software structure for running the simulation with robots. I organized the launch architecture around:

- a world file;
- `robots.yaml`;
- URDF files;
- SDF robot models;
- dynamic spawning through `ros_gz_sim create`.

I also worked on the runtime flow that:

- loads the world;
- filters enabled robots;
- resolves the correct model and URDF for each robot;
- launches `robot_state_publisher`;
- generates bridge configurations for `ros_gz_bridge`.

For mobile robots, I added logic to rewrite TIAGo topics per robot instance so the architecture can support namespaced multi-robot execution instead of only one robot. This was a major architectural step because it transformed the simulation from a static world into a reusable runtime system.

Week 6

During Week 6, I focused on the hardest implementation part so far: the shuttle and switch system. I first tried to model the shuttle dynamically, but it did not work properly because the shuttle got stuck and the wheel/contact-point behavior between the shuttle and the rails was not yet solved.

Instead of remaining blocked on the dynamic model, I switched to a more useful kinematic implementation. I extracted points for the big loop and the small loop, exported them to Excel, created spline-based paths, and used a set-position method in code to move the shuttle along those paths.

This gave me the first working shuttle behavior, even though it still needed refinement. The important result of this week was that shuttle motion became visible, testable, and ready to evolve into the more structured system that was developed later.

Overall Progress After Six Weeks

After these first six weeks, the project had already achieved the following foundation:

- a reconstructed and corrected MFJA third-floor environment;
- a Gazebo-ready floor and room model structure;

- clearer repository organization and documentation;
- a practical ROS–Gazebo architecture for multi-robot simulation;
- namespaced support for multiple robot instances;
- the first working shuttle implementation based on kinematic motion.

In other words, these six weeks built the technical base that made the later Room 315 kinematic shuttle system possible.

Conclusion

The first six weeks were mainly about building the project backbone. The work started from repository setup and environment correction, then progressed toward software architecture and finally reached the first operational shuttle behavior. Although the dynamic shuttle approach was not yet successful, the decision to move to a kinematic implementation ensured steady progress and created a working basis for the next development stages.